Word2Vec Semantic Analysis and Network Textual Analysis

Course: Computational Content Analysis

Company: Cisco Systems, Inc.

CIK: 858877

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May 24, 2026

Data Description

This analysis is based on five annual 10-K filings submitted by Cisco Systems, Inc. (CIK: 858877) to the U.S. Securities and Exchange Commission. The filings cover fiscal years 2021 through 2025, each corresponding to Cisco's fiscal year ending in late July of the respective year. The five documents used are: 2021_10K.txt, 2022_10K.txt, 2023_10K.txt, 2024_10K.txt, and 2025_10K.txt. Each file was downloaded directly from the EDGAR database and contains the complete text of the primary 10-K document after HTML stripping. Each file is treated as a single document representing one fiscal year, producing a five-document corpus suitable for year-level comparison of semantic patterns in management disclosures. All subsequent analyses, including Word2Vec modelling, thematic clustering, and co-occurrence network construction, are performed on this corpus.

Part 1: Corpus Construction and Word2Vec Model

Question 1: Corpus Construction and Text Preprocessing

Vocabulary Summary

Metric	Value
Vocab before preprocessing	9457
Vocab after preprocessing	3682
Terms removed	5775
DTM documents	5
DTM unique terms	3682

Top 30 Most Frequent Terms

	Term	Frequency
1	product	2038
2	service	1462
3	customer	1454
4	financial	1287
5	revenue	1284
6	tax	1048
7	asset	1013
8	security	956
9	value	929
10	business	925
11	market	891
12	income	882
13	based	835
14	sale	820
15	term	816
16	cash	804
17	operating	795
18	investment	789
19	net	774
20	cisco	737
21	loss	726
22	share	690

	Term	Frequency
23	contract	688
24	change	679
25	stock	676
26	risk	661
27	cost	661
28	technology	649
29	future	631
30	fair	616

DTM Dimensions

The Document-Term Matrix contains 5 documents and 3,682 unique terms, where each row corresponds to one fiscal year and each column to one lemmatised token present in the cleaned corpus.

Word Cloud

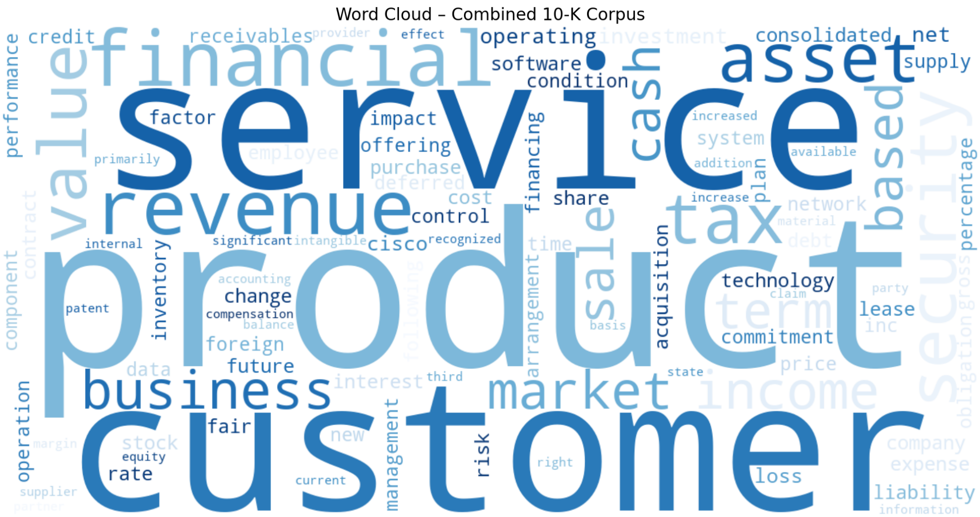

Figure 1: Word Cloud of Combined 10-K Corpus (post-preprocessing)

Interpretation

The top 30 most frequent terms in Cisco Systems' five-year 10-K corpus are dominated by core operational and financial vocabulary, which is consistent with the nature of SEC annual filings and the company's primary business activities. Terms such as product, service, customer, revenue, market, and business rank among the highest frequency words, confirming that Cisco's management communication across the five filings is centred on its networking and enterprise technology operations. This result aligns with expectations for a technology company whose revenue is driven by hardware products and subscription-based services.

Financial reporting vocabulary is well represented in the high-frequency terms. Words such as asset, income, cash, investment, net, loss, and tax reflect the accounting structure of the 10-K, which requires extensive disclosure of balance sheet items, income statement results, and tax obligations. The presence of these terms at high frequency indicates that financial condition reporting occupies a large proportion of the filing text, as required by SEC regulations.

Cisco-specific terms, particularly security and technology, appear prominently, reflecting the company's strategic focus on cybersecurity products and enterprise infrastructure. The term contract is also notable, pointing to Cisco's significant recurring revenue from multi-year software and service agreements. The appearance of risk and loss in the top 30 is consistent with mandatory risk factor disclosure sections present in every 10-K filing.

Overall, the top 30 terms are dominated by meaningful financial and business vocabulary. The expanded custom stopword list and lemmatisation approach have successfully removed generic administrative noise, leaving a corpus that is appropriate for the downstream Word2Vec and co-occurrence network analyses.

Part 2: Semantic Proximity and Thematic Clustering

Part 2 results will be added after semantic proximity and thematic clustering analysis.

Part 3: Co-occurrence Network and Structural Analysis

Part 3 results will be added after co-occurrence network, centrality, and community detection analysis.

Conclusion

As of the current stage of this report, the corpus construction and text preprocessing for Cisco Systems, Inc.'s five-year 10-K filings have been completed. The cleaned corpus consists of lemmatised tokens drawn from five annual reports spanning fiscal years 2021 to 2025, with domain-irrelevant and generic stopwords removed. The vocabulary was reduced from 9,457 raw terms to 3,682 unique lemmatised terms. The top 30 most frequent terms confirm that the corpus is dominated by meaningful financial and operational vocabulary, providing a reliable foundation for the subsequent analytical stages.

Subsequent sections will extend this foundation using a Word2Vec skip-gram model to capture semantic relationships between financial terms, k-means clustering to identify thematic vocabulary groupings, and co-occurrence network analysis to reveal structural patterns in Cisco's management language. These analyses will be completed and reported in Parts 2 and 3 of this document.